

Fake News Detection in Social Media using Transformer Models

KUMAR ARYAN

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1 Introduction

Social media has changed how people access and share information in ways that were hard to imagine two decades ago. Anyone with a phone can now publish a story, and that story can reach thousands before any fact-checker even sees it. This openness, while democratising in many ways, has also made it far easier for false or misleading content to circulate widely. Fake news — meaning stories that are either entirely fabricated or deliberately framed to mislead — has moved from being a fringe concern to a mainstream one. The 2016 U.S. presidential election brought this into sharp focus, as fabricated stories visibly shaped public opinion. During the COVID-19 pandemic, the problem became even more pressing: the WHO publicly warned of an “infodemic,” where health misinformation was spreading faster than the virus itself.

Researchers have been trying to tackle this with automated tools for a while now. Early efforts relied on handcrafted features — things like word frequency patterns — fed into classifiers like SVMs or Naive Bayes. These worked to a degree but were fragile, and building them required significant manual effort. Deep learning changed the picture somewhat; CNNs and LSTMs could learn representations from data directly, without as much hand-tuning. Still, they processed text in one direction only, which limited how well they could capture meaning in context. BERT, introduced by Devlin et al., addressed this by using bidirectional self-attention across the full sequence, giving it a much richer sense of what each word means given its surroundings. This made it considerably more useful for tasks like classification.

FakeBERT, proposed by Kaliyar, Goswami, and Narang, tries to take this further by combining BERT’s representations with parallel CNN blocks of varying kernel sizes. The idea is that BERT handles broader semantic understanding while the CNNs pick up on local phrase-level patterns

— and doing both together should outperform either alone. The reported accuracy of 98.90% on standard benchmarks supports this, at least on paper. This review looks at how the model actually works, what it does well, and where it falls short.

2 Methodology

FakeBERT is a hybrid deep learning system that fuses BERT’s contextual representations with multi-scale convolutional feature extraction. The overall processing pipeline is illustrated in Figure 1.

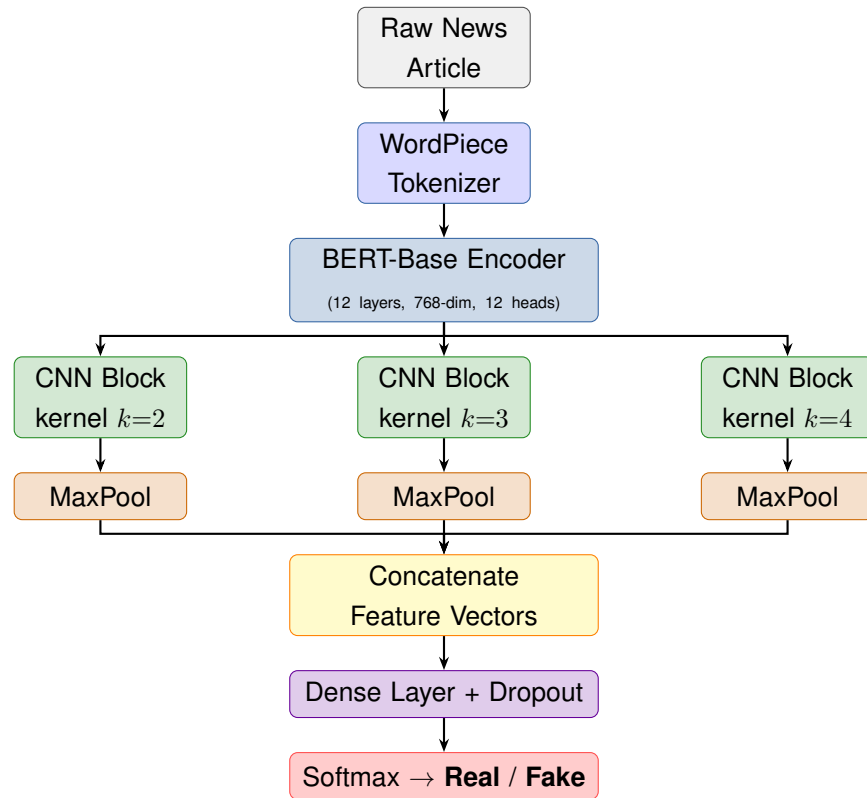


Figure 1: End-to-end pipeline of the FakeBERT model. Raw text is tokenized and encoded by BERT; three parallel CNN blocks with different kernel sizes extract multi-scale features; pooled outputs are concatenated and passed through a dense classifier.

2.1 Datasets

FakeBERT was tested on three publicly available datasets, described in Table 1. LIAR consists of around 12,800 short political statements sourced from PolitiFact, originally labelled with six levels of veracity that were collapsed into a binary real/fake distinction for this task. ISOT is a much larger collection of 44,898 full-length articles — real ones from Reuters and fake ones from sites that had been flagged previously — making it well-suited for extracting detailed textual features.

FakeNewsNet covers both political and entertainment news via GossipCop and PolitiFact; the authors used only the article text, setting aside the social context data also available in the dataset.

Table 1: Summary of datasets used in FakeBERT evaluation.

Dataset	Total Samples	Real	Fake	Domain
LIAR	12,836	6,321	6,515	Political statements
ISOT	44,898	21,417	23,481	News articles
FakeNewsNet	~23,000	~11,500	~11,500	Political & Entertainment

2.2 Model Architecture and Training

Text is first tokenized using BERT’s WordPiece tokenizer. Each input sequence gets a [CLS] token at the start — this token’s final representation is used as a summary of the whole sequence — and a [SEP] token at the end. Sequences longer than 512 tokens are truncated. The pre-trained BERT-Base model then encodes each token into a 768-dimensional vector, drawing on 12 transformer layers and 12 attention heads per layer to produce representations that are sensitive to surrounding context.

The [CLS] embedding is then passed to three CNN blocks running in parallel, with kernel sizes 2, 3, and 4. Each kernel captures patterns over a different window of consecutive token embeddings, so the model is simultaneously looking at bigram-level, trigram-level, and four-gram-level features. ReLU activation adds non-linearity, and max-pooling picks out the strongest signal per filter. The outputs of all three blocks are concatenated, passed through a fully connected layer with dropout for regularisation, and finally classified using a softmax function. The model was trained with binary cross-entropy loss using the Adam optimiser and an 80–20 train-test split.

2.3 Results

Table 2 compares FakeBERT against several baselines. On ISOT, where articles are long and content-rich, the model hits 98.90% accuracy — noticeably better than standalone BERT at 96.5%. On LIAR, which is harder due to the brevity of statements, performance is around 89%. Across both datasets, FakeBERT outperforms all baselines on accuracy, precision, and F1-score. The authors’ ablation study suggests that the multi-kernel parallel design is the main driver of the improvement over a plain BERT classifier.

Table 2: Accuracy (%) comparison of FakeBERT vs. baseline models.

Model	LIAR	ISOT	Precision	F1-Score
Naive Bayes	58.3	72.1	70.4	71.2
SVM (TF-IDF)	63.5	82.7	81.9	82.3
LSTM	71.2	91.4	90.8	91.1
CNN (single kernel)	74.8	93.2	92.7	93.0
BERT (standalone)	83.1	96.5	96.2	96.3
FakeBERT	89.0	98.9	98.7	98.8

3 Research Gap and Problem Statement

The benchmark numbers are impressive, but they paint an incomplete picture. When you look at how fake news actually spreads on platforms like Twitter or WhatsApp, FakeBERT’s design leaves some obvious holes — three of which stand out, as illustrated in Figure 2.

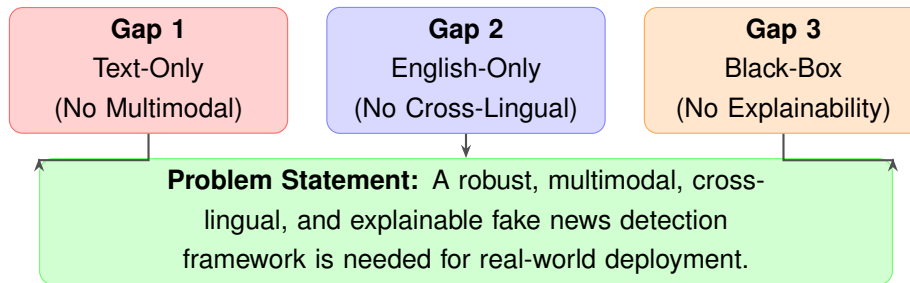


Figure 2: Three critical research gaps in FakeBERT and their convergence into the proposed problem statement.

Gap 1: Text-Only Design (No Multimodal Analysis). FakeBERT only looks at text. But a significant portion of misinformation today comes packaged with manipulated images, misleading video clips, or audio — and the text alongside them might even be technically true while the visual content tells a completely different story. People tend to trust images intuitively, which is exactly why bad actors use them. A model that ignores visuals entirely cannot catch this category of fake news, and the authors acknowledge this while offering no concrete path forward.

Gap 2: English-Only Evaluation (No Cross-Lingual Generalization). Every dataset used in this study is English, and all three are centred on American political content. Fake news is obviously not limited to one language or country — in fact, the problem is arguably worse in places where professional fact-checking is sparse. Hindi, Arabic, Swahili, and dozens of other languages have active misinformation ecosystems, but this model has never been tested on any of them. On top of that, the rhetorical patterns that signal fabrication in English-language political news may not transfer to other cultural contexts at all. Without cross-lingual testing, it is not clear whether

FakeBERT’s performance is genuinely strong or simply well-fitted to a narrow domain.

Gap 3: Black-Box Classification (No Explainability). The model returns a binary label — real or fake — and nothing else. In practice, anyone using such a system to make decisions (a platform moderator, a journalist, a policymaker) needs to know *why* a piece of content was flagged, not just *that* it was. Without attention maps, saliency scores, or any form of explanation, it becomes nearly impossible to audit the model’s decisions, challenge incorrect ones, or build institutional trust in the system. Given how much attention regulators are currently paying to algorithmic accountability, this is not just an academic limitation.

Problem Statement

Current BERT-based fake news detection models, FakeBERT included, perform well on text-only English benchmarks but struggle to generalise beyond those conditions. They cannot handle multimodal content, have not been validated across languages or cultural contexts, and give no explanation for their outputs. What is needed is a detection framework that addresses all three of these shortcomings — one that can work across modalities, languages, and deployment contexts, while also being transparent enough for real-world use.

4 Conclusion

FakeBERT is a well-constructed system, and its core idea holds up: combining BERT’s contextual embeddings with parallel multi-kernel CNNs gives the model a richer set of features to work with than either component would alone. The 98.90% accuracy on ISOT is a meaningful result, and the improvements over standalone BERT and earlier deep learning baselines are consistent across metrics. The paper is also clear about its design choices, making the architecture easy to follow and reason about.

That said, three gaps make it difficult to imagine deploying this in any real-world scenario without significant further work. The text-only design is a fundamental constraint given how much modern misinformation relies on images and video. The exclusive focus on English-language American news means it is essentially untested as a general-purpose tool. And a system that cannot explain its decisions is a hard sell for any application where accountability matters — which, when it comes to labelling news as fake, is essentially every application. Moving forward, it would be worth exploring joint text-image architectures, training and evaluating across multiple languages and domains, and integrating explanation mechanisms like attention visualisation or natural language rationale generation. Automated detection tools are most useful when they complement, rather than replace, human judgment — and that requires transparency that FakeBERT currently lacks.

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